



[Sign up for this introductory AI course to gain practical, industry-ready AI skills you'll need to increase productivity, innovate, create, and succeed. These skills include understanding how modern AI systems work, learning what is AI and what is not, using leading AI tools for research, creating study guides for deeper subject matter understanding, evaluating AI's strengths and limitations, and applying ethical and faith-informed principles to real-world AI challenges.]

How to get the most out of [A Catholic Introduction to AI, Part 1](#):

- Watch the introductory videos for the course.
- Read the assignments before class meetings.
- Have a notebook ready and available for class notes during each live or recorded session.
- Do the assignments, quizzes, and labs assigned for that week.

Once the course is completed to the parent's and instructor's satisfaction, there is a Certificate of Completion at the end to be filled in for your records.

Total Classes: 13 Weeks

Class Day and Dates: Tuesdays, September 8 to December 15, 2026. (No class Oct. 13 for fall break, and Dec. 8 for the Holy Day)

Starting Time: 4:00 PM Eastern (3:00 Central; 2:00 Mountain; 1:00 Pacific)

Duration: 60 minutes

Suggested Grade Level: High School

Suggested Credit: 2 full semesters Computer Science

Instructor: Mr. Daniel Goddu, BS-CS

Textbook: None; all materials provided with the course with no extra cost.

Prerequisites: One of the following from Homeschool Connections or an equivalent programming course (Python preferred) outside of Homeschool Connections:

- Introduction to Computer Science, Part 1 or Part 2
- Introduction to Computer Science Using SNAP!
- Computer Programming 101 with Python
- C# Programming with Unity 101

Course Requirements:

This course uses the following AI-based tools:

- **NotebookLM** is an AI research and study tool developed by Google that helps users understand and work with their own documents. Instead of answering questions from the general internet, it analyzes sources you upload—such as PDFs, Google Docs, or notes—and generates



summaries, explanations, study guides, and even audio-style discussions based only on those materials. This makes it especially useful for research, learning, and organizing information because the AI stays grounded in the sources you provide. **A Google account is required.**

- **Jupyter Notebook** is an open-source interactive computing environment where users can write code, display results, add explanations, and create visualizations all in one document. It is widely used in data science, machine learning, and AI because it allows programmers to run code step-by-step while documenting their work. This makes it especially useful for learning, experimentation, and sharing reproducible research.

Jupyter Notebooks are best created and executed using either **Visual Studio Code** (software download required, no account required) or **Google Colab** (Google account required). Refer to the description of these platforms below (choose either VSC or Colab).

- **Visual Studio Code (VSC):** VSC is an integrated development environment (IDE) that aids in the writing, debugging, and execution of a large range of computer programs. It also can handle Jupyter Notebooks. Students with computer programming experience and who are familiar with VSC will find using VSC for Jupyter Notebooks quite easy. VSC software is free but it must be installed on the student's computer or laptop.
- **Google Colab:** (short for *Google Colaboratory*) is a free cloud-based environment from Google that lets users write and run Python code in a web browser using Jupyter Notebooks. Because it runs in the cloud and can provide access to GPUs, students can build and test AI models without installing software on their own computers. **A Google account is required.**

Access to the following open-source Large Language Models (LLMs):

- **Ollama (<https://ollama.com/>):** Ollama (OLLama) is an **open-source**, large-scale language model developed by Alibaba Cloud. It can generate natural-sounding text responses based on the prompt provided, making it capable of understanding and engaging with complex topics across multiple domains. **An Ollama free-usage account is required.**

Access to **one** of the following cloud-based proprietary AI models:

- **ChatGPT (<https://chatgpt.com/>):** an AI assistant that explains ideas, answers questions, researches information, and helps with writing, learning, planning, and problem-solving across many topics (from Chat GPT). **No account registration required.**
- **Google Gemini (<https://gemini.google.com/app>):** I am Gemini 3 Flash, an adaptive AI collaborator. I'm here to help you create text, images, music, and video with clarity and wit (from Gemini). **A Google account is required.**
- **Claude (<https://claude.ai/login>):** an AI assistant made by Anthropic. I can answer questions, write, analyze, code, and help with a wide range of tasks (from Claude). **Account registration is required.**



Course Description:

This course provides a hands-on and critical introduction to Artificial Intelligence, covering its history, foundational concepts, societal impact, and ethical concerns. Part One introduces students to the foundations of artificial intelligence, beginning with what AI is and how it developed from early automation to modern machine learning. Students explore key concepts such as neural networks, deep learning, and large language models, while also examining how AI is used across industries like healthcare, finance, gaming, and the arts. The semester also emphasizes critical thinking about the social and ethical implications of AI, including issues of bias, privacy, surveillance, and the future of work. Students will gain hands-on experience building and working with modern AI tools. By the end of Part One, students will have a solid conceptual understanding of AI and be prepared to gain more hands-on work with modern AI tools in Part Two.

Part Two focuses on more hands-on experience building and working with modern AI tools. Students learn how large language models operate in practice by running open-source models locally using tools such as Ollama and interacting with them through interfaces like Open WebUI. They explore prompt engineering, system prompts, tools, and functions, and experiment with AI-powered automation using platforms such as Make.com (or equivalent). The semester concludes with projects in which students design simple AI assistants or agents, giving them practical experience applying AI responsibly and creatively.

Students will engage in **interactive discussions** and **hands-on AI experiments**. The class will encourage **critical thinking about AI's benefits, risks, and ethics**. While students will not be required to write any computer programs, a course in computer programming is required as a prerequisite. A recorded tutorial on Python and SNAP! are part of this course to bring the student up to a level of the required proficiency to be successful for the course. Programming examples are provided. The student will learn and be evaluated on what AI objective the code achieves.

This course integrates faith and technology to explore faith, ethics, and AI advancements together. The Catholic Church views science to explore and comprehend the natural world created by God. Technology is encouraged when it respects ethical principles and upholds the dignity of human life. Catholics recognize the benefits of technological advancements for improving human life and society. Moral discernment is crucial to guide technology to serve humanity ethically and avoid harm or injustice.

This course will not teach students how to exploit AI concepts that violate academic integrity. Its main goal is to introduce the student to AI concepts and the technology that makes it happen. Over the course of the topics, students will understand the so-called *truth* that AI provides and the truth that it doesn't provide. And it will cut through the fantasy and hype so often attributed to AI.

Semester 1: AI Overview

Class 1: Course Overview & What is AI?

This class introduces students to the basic concepts of Artificial Intelligence. They will learn the difference between AI, Machine Learning, and Deep Learning, as well as explore real-world AI applications in personal assistants (Siri, Alexa), self-driving cars, and recommendation systems (Netflix,



YouTube). The class will also discuss common misconceptions about AI and the different types of AI (narrow, general, and superintelligence).

Class 2: The Early History of AI & Automation

Students will explore the history of automation before computers, including the Jacquard Loom and Henry Ford's assembly line. The discussion will then transition to early AI theories, such as Alan Turing's work and the creation of early automated systems. We'll also examine how automation in factories led to the first wave of robotic manufacturing.

Class 3: AI in the 20th Century

This class covers the development of early AI programs, including ELIZA (one of the first chatbots) and expert systems used in decision-making. Students will learn about AI's role in space exploration, the military, and industry. We will also discuss the "AI Winter" periods when enthusiasm and funding for AI declined due to technological limitations.

Class 4: Machine Learning & Neural Networks

Students will be introduced to Machine Learning, how AI learns from data, and how neural networks function similarly to the human brain. Concepts like supervised vs. unsupervised learning will be explained. Students will participate in a hands-on activity, using AI tools like Google Teachable Machine to train a simple model and see how it learns patterns.

Class 5: Deep Learning & LLMs

This class explores deep learning, the foundation of powerful AI models like ChatGPT and Bard. Students will learn how LLMs process language and generate responses. Hands-on experiments will involve interacting with LLMs, testing their strengths and weaknesses, and discussing how these systems are trained. The class will also address common misconceptions about AI chatbots.

Class 6: Ethics of AI – Bias & Fairness

Students will learn about how AI can reflect biases found in its training data, leading to discrimination in hiring, policing, and facial recognition. Case studies will highlight real-world examples of AI bias and its consequences. The class will explore ways to make AI more ethical and fairer, followed by a discussion on who is responsible for AI's actions.



Class 7: AI in Industry – Healthcare, Finance, and More

Students will examine how AI is transforming industries like healthcare (medical diagnosis), finance (fraud detection), and agriculture (crop monitoring). We will analyze case studies of AI improving efficiency while also discussing potential risks (such as job loss and decision-making bias). Groups will research and present how AI is used in a specific industry of their choice.

Class 8: AI in Gaming & Entertainment

This class covers AI's role in gaming (e.g., NPC behavior, procedural content generation) and entertainment (AI-generated music, deepfake videos). Students will explore how AI tools are used to create digital art and animation. A hands-on activity will allow them to generate AI-created music or images. We will discuss whether AI-created content can be considered "real" creativity.

Class 9: AI and the Future of Work

This session focuses on AI's impact on employment. We will discuss the types of jobs AI can replace (factory work, data entry) versus those it will enhance (creative fields, engineering). Students will analyze reports on the future job market and predict which careers AI might create. The class ends with a debate on whether AI will create more jobs than it eliminates.

Class 10: The Morality of AI & Autonomous Systems

Can AI be programmed to be ethical? This class will discuss moral dilemmas involving AI, such as its role in the military (autonomous drones) and medical decision-making (AI doctors). We will examine real-world ethical debates, including the controversial topic of AI-powered weapons. A class debate will tackle whether AI should be allowed to make life-and-death decisions.

Class 11: AI & Privacy – Risks & Security

AI-powered surveillance, facial recognition, and data collection present major privacy concerns. Students will explore how AI is used to track online behavior and influence decision-making. We will examine the dangers of AI-generated misinformation (deepfakes, social media manipulation). A hands-on activity will involve testing AI detection tools for spotting deepfakes.

Class 12: AI Policy, Laws & the Future of Regulation

With AI's rapid advancements, governments are scrambling to create regulations. Students will explore different approaches to AI governance, from strict EU regulations to the more flexible U.S. approach. We will discuss the ethical responsibilities of tech companies and policymakers. A class discussion will focus on whether AI research should be slowed down or left unrestricted.



Class 13: The Future of AI & Final Project

The final class looks at where AI is headed, including the potential development of Artificial General Intelligence (AGI). Students will present their final projects on AI's future in different fields, exploring both its potential benefits and risks. The course will conclude with a reflection on what students have learned and how they see AI shaping the world.

Assessment Overview

- **Short essays – 30%**
- **Quizzes – 10%**
- **Labs – 20%**
- **Final Project – 40%**

Bibliography

- [1] Russell, Stuart, and Peter Norvig. Artificial Intelligence: A Modern Approach. Pearson, 2020.
- [2] Muelier, John Paul, Luca Massaron, and Stephanie Diamond. Artificial Intelligence for Dummies. John Wiley & Sons, Inc. 2025.
- [3] Poole, David L, and Alan K Mackworth. Artificial Intelligence 3E: Foundations of Computational Agents. Cambridge University Press, 2023. <https://artint.info/>
- [4] Bulloch, Seth. Artificial Intelligence: Teaching Artificial Intelligence to High School Students. Kindle Ed.